

Introduction to Real Analysis – Homework 4

Measurable functions and simple integrals

Notation. m is Lebesgue measure and $I(s)$ is the integral of an unsigned simple function. Use $0 \cdot \infty = \infty \cdot 0 = 0$. All measurability is Lebesgue measurability.

1. **Large values on a set of finite measure.** Let $f : \mathbb{R}^d \rightarrow [0, \infty]$ be measurable and let $E \subset \mathbb{R}^d$ be measurable with $m(E) < \infty$.

(a) Prove that

$$\lim_{n \rightarrow \infty} m(E \cap \{f \geq n\}) = m(E \cap \{f = \infty\}).$$

Deduce that the limit is zero if and only if f is finite a.e. on E .

- (b) Give an example with $m(E) = \infty$ and f finite everywhere for which $m(E \cap \{f \geq n\})$ does not tend to zero.

2. **Operations on measurable functions.** Let $f, g : \mathbb{R}^d \rightarrow [0, \infty]$ be measurable.

(a) Prove that $\max\{f, g\}$ and $\min\{f, g\}$ are measurable.

- (b) Prove that $f + g$ is measurable. *Hint:* express its superlevel sets using a countable union indexed by rational numbers.

3. **Simple integrals and null sets.**

(a) Put $D = \mathbb{Q} \cap [0, 2)$ and

$$s = \mathbf{1}_{[0,1)} + 2\mathbf{1}_{[1/2,2)} + \infty \mathbf{1}_D.$$

Compute $I(s)$.

- (b) If unsigned simple functions $s, t : \mathbb{R}^d \rightarrow [0, \infty]$ agree a.e., prove that $I(s) = I(t)$, allowing the common integral to be infinite.

4. **Level-set estimates.** Let $s : \mathbb{R}^d \rightarrow [0, \infty]$ be an unsigned simple function.

(a) For every real $\lambda > 0$, prove that

$$m(\{s \geq \lambda\}) \leq \frac{I(s)}{\lambda}.$$

- (b) Deduce that $I(s) = 0$ if and only if $s = 0$ a.e., and that $I(s) < \infty$ implies $s < \infty$ a.e.